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ENGINEERING MATHEMATICS II

UNIT 2 & LP8 – MECHANICAL VIBRATIONS WITH AND WITHOUT DAMPING

Mechanical Vibrations

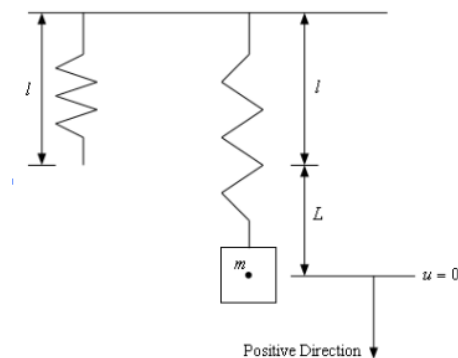
It's now time to take a look at an application of second order differential equations.

We're going to take a look at mechanical vibrations. In particular we are going to look at a mass that is hanging from a spring.

Vibrations can occur in pretty much all branches of engineering and so what we're going to be doing here can be easily adapted to other situations, usually with just a change in notation.

Let's get the situation setup. We are going to start with a spring of length l , called the natural length, and we're going to hook an object with mass m up to it. When the object is attached to the spring the spring will stretch a length of L . We will call the equilibrium position the position of the center of gravity for the object as it hangs on the spring with no movement.

Below is sketch of the spring with and without the object attached to it.



As denoted in the sketch we are going to assume that all forces, velocities, and displacements in the downward direction will be positive. All forces, velocities, and displacements in the upward direction will be negative.

Also, as shown in the sketch above, we will measure all displacement of the mass from its equilibrium position. Therefore, the $u = 0$ position will correspond to the center of gravity for the mass as it hangs on the spring and is at rest (i.e. no movement).

The General Differential Equation Modelling the System

To derive the equation governing the motion of a spring-mass-damper system, we apply Newton's second law of motion, which relates the net force acting on the mass to its acceleration. The primary forces acting on the mass in a spring-mass system include:

- Force due to gravity $F_g = mg$ acting downward.
- Restorative Force of the spring $F_s = -k(l + y)$, where k is the spring constant. This force is governed by Hooke's law and is typically proportional to the displacement from the spring's natural length (l_0) and opposite in direction.
- Damping Force $F_d = -cy'$, where c is the damping coefficient. If present, the damping force is proportional to the velocity of the mass and acting in the opposite direction of motion.
- External Force $F(t)$. It includes any external force acting on the system, which might be periodic or random, leading to forced vibrations.

Depending on which forces act on the system, there are several special cases:

- **Free Undamped Vibration** ($c = 0$, $F(t) = 0$): The simplest form of vibration occurs when there is no damping and no external force. The system oscillates at its natural frequency, determined by the mass and spring constant.
- **Free Damped Vibration** ($c > 0$, $F(t) = 0$): When damping is present but there is no external force, the system experiences damped vibrations leading to a gradual decrease in oscillation amplitude over time. The nature of the damping (underdamped, critically damped, or overdamped) depends on the values of m , c , and k .
- **Forced Undamped Vibration** ($c = 0$, $F(t) \neq 0$): When an external force acts on the system, the system experiences forced vibrations. If the frequency of the external force is close to the system's natural frequency, resonance can occur, leading to large amplitude oscillations.
- **Forced Damped Vibration** ($c > 0$, $F(t) \neq 0$): This is the most general case, combining the effects of damping and external forcing, leading to complex oscillatory behavior.

Considerations for Units and Phase Angle

- **Units:** When working with the acceleration due to gravity or any other physical quantity, it is important to use consistent units throughout the calculation. In the metric system, g is typically given as 9.81 m/s^2 , and lengths should be in meters with mass in kilograms. In the Imperial system, g is about 32 ft/s^2 , with lengths in feet and mass in slugs.
- **Phase Angle Uniqueness:** There are infinitely many phase angles that satisfy the trigonometric equations due to their periodic nature. However, selecting ϕ in the interval $[-\pi, \pi)$ ensures a unique solution within one complete cycle. The signs of c_1 and c_2 determine the quadrant in which ϕ lies.
 - if $c_1, c_2 > 0$, ϕ is in the first quadrant
 - If $c_1 < 0$, $c_2 > 0$, ϕ is in the second quadrant.
 - if $c_1, c_2 < 0$, ϕ is in the third quadrant.
 - If $c_1 > 0$, $c_2 < 0$, ϕ is in the fourth quadrant.

Free Damped Vibration

In free, damped vibration, there is no external force ($F(t) = 0$). As such, Equation 3.8.1 simplifies to a homogeneous second-order linear differential equation.

$$my'' + cy' + ky = 0 \quad (3.8.5)$$

This equation is a homogeneous second-order linear differential equation. By solving the characteristic equation $mr^2 + cr + k = 0$ using the quadratic formula, we find the roots

$$r_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

Depending on the discriminant $c^2 - 4mk$, we encounter three types of motion:

1. Critically damped ($c^2 - 4mk = 0$)

In this case, there is a repeated root $r = -\frac{c}{2m}$, and thus the general solution to Equation 3.8.5 becomes

$$y(t) = c_1 e^{-\frac{c}{2m}t} + c_2 t e^{-\frac{c}{2m}t} \quad (3.8.6)$$

The motion in this case is said to be critically damped as the damping is just enough to prevent oscillation. This level of damping is achieved when the damping coefficient c .

$$c^2 - 4mk = 0$$

$$c^2 = 4mk$$

$$c = 2\sqrt{mk}$$

$2\sqrt{mk}$ is called the critical damping coefficient and is denoted by C_{cr} .

It is important to note that as time progresses ($t \rightarrow \infty$), the displacement $y(t)$ approaches zero, indicating that the system smoothly and quickly settles to its equilibrium position without oscillation and overshooting the equilibrium position similar to how shock absorbers work in automotive suspension systems.

2. Overdamped ($c^2 - 4mk > 0$)

In this case, there are two real distinct roots $r_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$, where both roots are negative. The general solution to Equation 3.8.5 becomes

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t} \quad (3.8.7)$$

Given both r_1 and r_2 are negative, as time progresses ($t \rightarrow \infty$), the displacement $y(t)$ approaches zero and the system gradually returns to equilibrium without oscillating. Overdamped conditions arise when $c > c_{cr}$, typically desired in systems where overshooting the equilibrium position could be harmful or undesirable, like in heavy machinery. Overdamped systems return to equilibrium slower than critically damped systems. This slower response is due to the higher damping force applied, which prevents oscillation but also resists motion, causing a sluggish return.

3. Underdamped ($c^2 - 4mk < 0$)

In this case, the roots of the characteristic equation are complex conjugates given by

$$r_{1,2} = -\frac{c}{2m} \pm \frac{\sqrt{4mk - c^2}}{2m}i = -\frac{c}{2m} \pm \omega_1 i$$

Thus the solution to the differential Equation 3.8.5 is

$$y(t) = e^{-\frac{c}{2m}t} (c_1 \cos(\omega_1 t) + c_2 \sin(\omega_1 t)) \quad (3.8.8)$$

The term $\omega_1 = \frac{\sqrt{4mk - c^2}}{2m}$ is related to the frequency of oscillation. Similar to the harmonic motion, we can derive the amplitude-phase form of the equation of motion.

$$y(t) = R e^{-\frac{c}{2m}t} \cos(\omega_1 t - \phi) \quad (3.8.9)$$

Here again

$$R = \sqrt{c_1^2 + c_2^2}, \quad \cos(\phi) = \frac{c_1}{R}, \quad \text{and} \quad \sin(\phi) = \frac{c_2}{R}$$

An underdamped system is characterized by a damping coefficient $c < c_{cr}$. In this scenario, the damping is insufficient to halt oscillations, causing the system to exhibit oscillatory behavior around the equilibrium position. The amplitude of these oscillations diminishes over time, represented by the time-varying term $Re^{-\frac{c}{2m}t}$. As the exponent $-\frac{c}{2m}$ is always negative, the displacement $y(t)$ gradually approaches zero as time progresses ($t \rightarrow \infty$). This results in a bouncy system response to any disturbances.

Such behavior is often preferred in various applications. In musical instruments, for example, the underdamped vibrations of strings or membranes contribute to a sustained, resonant sound. Similarly, seismic dampers in buildings employ a controlled underdamped response to safely dissipate energy from earthquakes, allowing structures to sway and reduce stress without collapsing.

Example 1

A 1-kg mass is attached to a spring with a stiffness of 64 N/m and a dashpot with a damping constant 16 N.s/m. The object is compressed 20 cm above its equilibrium and released with an initial upward velocity of 2 m/s. Find the displacement of the object as a function of time.

Given information:

- Mass of the object: $m = 1 \text{ kg}$
- Damping constant: $c = 16 \text{ N} \cdot \text{s/m}$
- Spring constant: $k = 64 \text{ N/m}$
- Initial displacement (upward): $y_0(0) = 20 \text{ cm} = 0.2 \text{ m}$
- Initial velocity (upward): $y'_0(0) = 2 \text{ m/s}$

The initial value problem for this system is

$$y'' + 16y' + 64y = 0, \quad y_0(0) = 0.2, \quad y'_0(0) = 2$$

Before solving the IVP, we can calculate the critical damping coefficient to determine the type of damping.

$$c_{cr} = 2\sqrt{1(64)} = 16$$

The damping coefficient equals the critical damping coefficient ($c = c_{cr} = 16$), and therefore, the system is critically damped.

Finding the general solution:

The general solution for a critically damped system is given by Equation 3.8.6.

$$y(t) = c_1 e^{-\frac{c}{2m}t} + c_2 t e^{-\frac{c}{2m}t}$$

$$y(t) = c_1 e^{-8t} + c_2 t e^{-8t}$$

Applying the initial conditions:

$$y_0(0) = 0.2$$

$$c_1 e^0 + 0 = 0.2$$

$$c_1 = 0.2$$

$$y'(t) = -8c_1 e^{-8t} + c_2 e^{-8t}(1 - 8t)$$

$$y'_0(0) = 2$$

$$-8c_1 e^0 + c_2 e^0(1) = 2$$

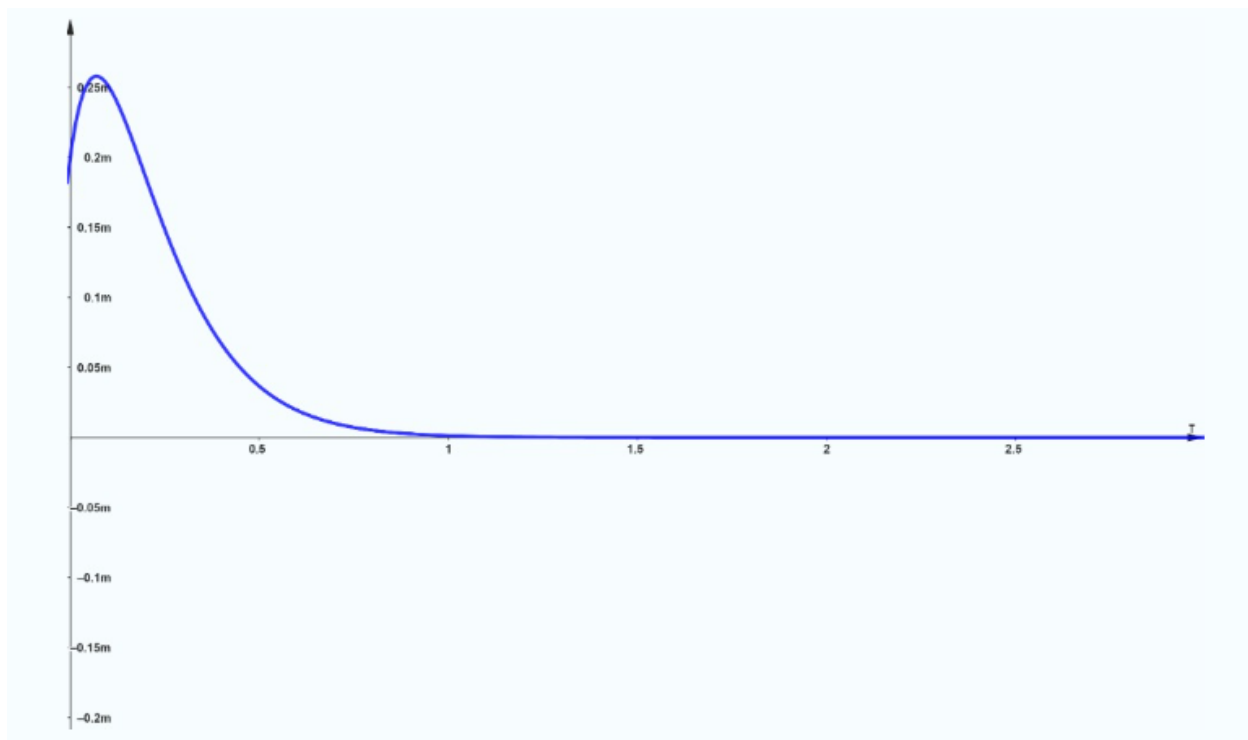
$$-8(0.2) + c_2 = 2$$

$$c_2 = 3.6$$

The equation of the object's displacement is then

$$y(t) = 0.2e^{-8t} + 3.6te^{-8t}$$

The graph of the displacement is shown for the first 3 seconds. As expected, the system smoothly and quickly returns to its equilibrium position without oscillation.



Practice problem

A 16 kg object is attached to a spring with spring constant 256 kg/s^2 . It is also attached to a dashpot with damping constant $c = 128 \text{ kg/s}$. The object is pulled down 0.2 m and released with an initial upward velocity of 2 m/s . Find the displacement of the object. Assume displacement and velocity are positive upward.

Example 2

A 1-kg mass is attached to a spring with a stiffness of 64 N/m and a dashpot with a damping constant $34 \text{ N}\cdot\text{s/m}$. The object is compressed 20 cm above its equilibrium and released with an initial upward velocity of 2 m/s . Find the displacement of the object as a function of time.

Given information:

Mass of the object: $m=1 \text{ kg}$

Damping constant: $c=34 \text{ N}\cdot\text{s/m}$

Spring constant: $k=64 \text{ N/m}$

Initial displacement (upward): $y_0(0)=20 \text{ cm}= 0.2 \text{ m}$

Initial velocity (upward): $y'(0)=2 \text{ m/s}$

The initial value problem for this system is

$$y''+34y'+64y=0, y_0(0)=0.2 \quad y'_0(0)=2$$

Finding the general solution:

The characteristic equation for the differential equation has two distinct real roots.

$$r_{1,2} = \frac{-34 \pm \sqrt{34^2 - 4(1)(64)}}{2(1)}$$

$$r_1 = -2, \quad r_2 = -32$$

The general solution to an overdamped system is given by Equation [3.8.7](#).

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

$$y(t) = c_1 e^{-2t} + c_2 e^{-32t}$$

Applying the initial conditions:

$$y_0(0) = 0.2$$

$$c_1 e^0 + c_2 e^0 = 0.2$$

$$c_1 + c_2 = 0.2$$

$$y'(t) = -2c_1 e^{-2t} - 32c_2 e^{-32t}$$

$$y'_0(0) = 2$$

$$-2c_1e^0 - 32c_2e^0 = 2$$

$$-2c_1 - 32c_2 = 2$$

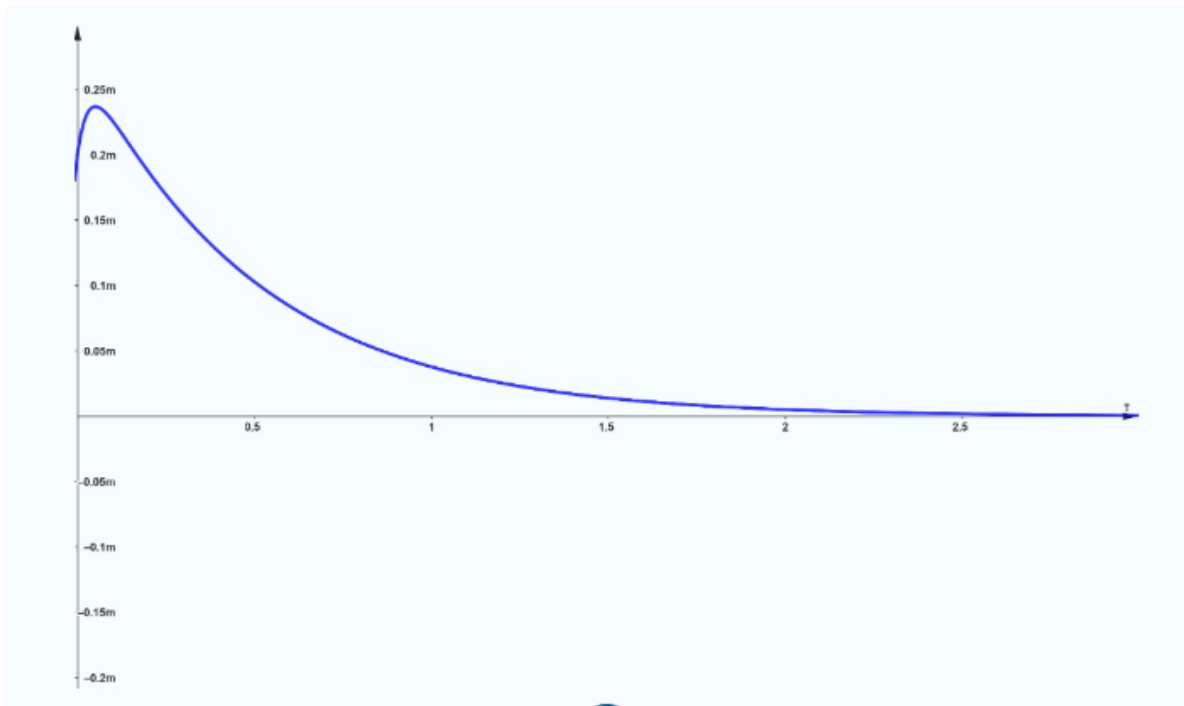
Solving the system for constants c_1 and c_2 yields

$$c_1 = 0.28, c_2 = -0.08$$

The equation of the object's displacement is then

$$y(t) = 0.28e^{-2t} - 0.08e^{-32t}$$

The below graph displays the displacement for the first 3 seconds. It confirms that the system gradually returns to its equilibrium position smoothly and without any oscillation. When compared to the critically damped system in Example 3.8.2, this overdamped system takes a longer time to settle down. This slower behavior underscores that the increased damping force in the overdamped system delays the return to equilibrium.



Example 3

A 1-kg mass is attached to a string with a stiffness of 64 N/m and a dashpot with a damping constant 4 N.s/m. The object is compressed 20 cm above its equilibrium and released with an initial upward velocity of 2 m/s. Find the displacement of the object as a function of time.

Given information:

Mass of the object: $m=1$ kg

Damping constant: $c=4$ N·s/m

Spring constant: $k=64$ N/m

Initial displacement (upward): $y_0(0)=20$ cm= 0.2 m

Initial velocity (upward): $y'(0)=2$ m/s

The initial value problem for this system is

$$y''+4y'+64y=0, y_0(0)=0.2 \quad y'_0(0)=2$$

Finding the general solution:

The characteristic equation for the differential equation has complex conjugates.

$$r_{1,2} = -\frac{4}{2(1)} \pm \frac{\sqrt{4(1)(64) - 4^2}}{2(1)}i = -2 \pm 2\sqrt{15}i$$

The general solution to an underdamped system is given by Equation 3.8.8.

$$y(t) = e^{-2t}(c_1 \cos(2\sqrt{15}t) + c_2 \sin(2\sqrt{15}t))$$

Applying the initial conditions:

$$y_0(0) = 0.2$$

$$e^0(c_1 \cos(0) + c_2 \sin(0)) = 0.2$$

$$c_1 = 0.2$$

$$y'(t) = e^{-2t}(-2c_1 \cos(2\sqrt{15}t) - 2c_2 \sin(2\sqrt{15}t) - 2\sqrt{15}c_1 \sin(2\sqrt{15}t) + 2\sqrt{15}c_2 \cos(2\sqrt{15}t))$$

$$y'_0(0) = 2$$

$$-2c_1 + 2\sqrt{15}c_2 = 2$$

$$-2(0.2) + 2\sqrt{15}c_2 = 2$$

$$c_2 = \frac{2\sqrt{15}}{25}$$

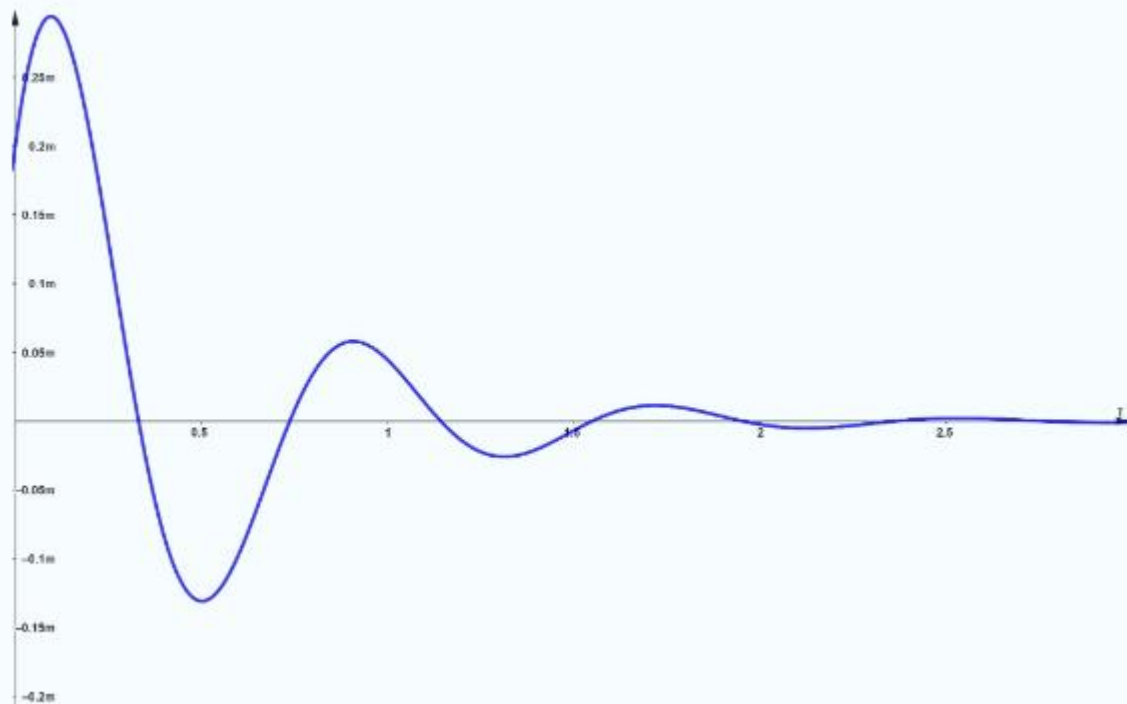
The equation of the object's displacement is then

$$y(t) = e^{-2t} \left(0.2 \cos(2\sqrt{15}t) + \frac{2\sqrt{15}}{25} \sin(2\sqrt{15}t) \right)$$

The amplitude-phase form of the equation is

$$y(t) = \frac{\sqrt{85}}{25} e^{-2t} \cos(2\sqrt{15}t - 0.9976)$$

The graph illustrates the displacement of the system over the initial 3 seconds. This underdamped system lacks enough damping to stop oscillations, leading to a pattern of diminishing swings around the equilibrium position. These oscillations decrease in amplitude for approximately 2 seconds before the system finally settles at the equilibrium position.



Practice problem:

1. A 16 kg object is attached to a spring with spring constant 12 kg/s^2 . It is also attached to a dashpot with damping constant 11 kg/s . The object is initially displaced 44 m above equilibrium and released. Find its displacement for $t > 0$.

2. A 99 kg object is attached to a spring with spring constant 288 kg/s^2 . It is also attached to a dashpot with damping constant $c = 108 \text{ kg/s}$. The object is pushed upwards from equilibrium with velocity 2 m/s . Find its displacement for $t > 0$.

Discussion Questions:

Try to answer the following questions

1. What is Mechanical Vibrations?
2. What is damping?
3. What is critical damping, over damping and under damping?
4. Find the displacement of the string with and without damping.